

Quivers

Leopold Wuhan Zhou
School of EECS, Peking University

July 23, 2026

Abstract

This note is based on the lecture by Victor Ginzburg at the University of Chicago in summer 2026. We introduce quivers and their finite-dimensional representations, describe representation spaces and the base-change action, state the finite-type form of Gabriel's theorem, and explain the Tits-form argument showing that finite representation type forces the underlying graph to be a simply laced Dynkin diagram.

Contents

1	Quivers and Their Representations	2
2	Dynkin Diagrams and Gabriel's Theorem	3
3	Representation Spaces and Base Change	3
4	A One-Parameter Family for $\tilde{D}_4$	4
5	The Tits Form and the Easy Direction of Gabriel's Theorem	5

1 Quivers and Their Representations

Throughout, let k be a field.

Definition 1.1. A *quiver* is a directed graph

$$Q = (I, E, s, t),$$

where I is the set of vertices, E is the set of arrows, and

$$s, t : E \longrightarrow I$$

assign to each arrow its source and target. An arrow $a \in E$ with $s(a) = i$ and $t(a) = j$ is written

$$a : i \longrightarrow j.$$

Unless stated otherwise, all quivers in these notes are finite.

Definition 1.2. A finite-dimensional representation ρ of a quiver Q over k consists of

- (i) a finite-dimensional k -vector space V_i of dimension $V_i = d_i$ for every vertex $i \in I$;
- (ii) a k -linear map

$$\rho(a) : V_i \longrightarrow V_j$$

for every arrow $a : i \rightarrow j$.

The vector

$$d = \dim V = (d_i)_{i \in I} \in \mathbb{Z}_{\geq 0}^I$$

is called the *dimension vector* of V .

Definition 1.3. Let V and V' be representations of Q . A *morphism*

$$f : V \longrightarrow V'$$

is a collection of linear maps $f_i : V_i \rightarrow V'_i$ such that, for every arrow $a : i \rightarrow j$,

$$f_j \circ \rho(a) = \rho'(a) \circ f_i. \tag{1.1}$$

Equivalently, the following square commutes:

$$\begin{array}{ccc} V_i & \xrightarrow{\rho(a)} & V_j \\ f_i \downarrow & & \downarrow f_j \\ V'_i & \xrightarrow{\rho'(a)} & V'_j \end{array}$$

The morphism f is an *isomorphism* if and only if every f_i is invertible.

Example 1.4. Consider the quiver



A representation is exactly a linear map $A : V \rightarrow V$. If another representation is given by $B : V' \rightarrow V'$, then a morphism from A to B is a pair (f_1, f_2) satisfying

$$f_2 A = B f_1.$$

Thus two representations are isomorphic precisely when their corresponding matrices are related by independent changes of bases in the domain and codomain.

2 Dynkin Diagrams and Gabriel's Theorem

The connected simply laced Dynkin diagrams are of types A_n , D_n , E_6 , E_7 , and E_8 .

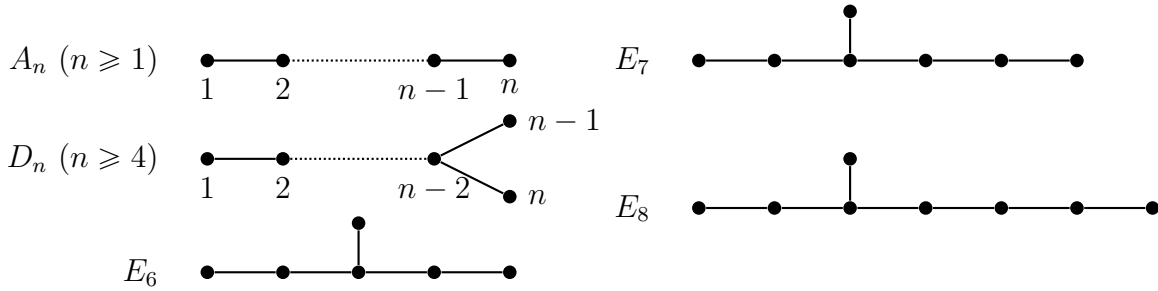


Figure 1: The connected simply laced Dynkin diagrams.

Theorem 2.1 (Gabriel). *Let k be algebraically closed and let Q be a connected quiver. The following are equivalent:*

- (1) *the underlying unoriented graph of Q is one of the Dynkin diagrams A_n , D_n , E_6 , E_7 , or E_8 ;*
- (2) *for every dimension vector $\mathbf{d}$, there are only finitely many isomorphism classes of representations of Q of dimension vector $\mathbf{d}$.*

Remark 2.2. The same simply laced Dynkin diagrams also appear in the classification of finite-dimensional complex simple Lie algebras and, through the McKay correspondence, in the classification of finite subgroups of $\mathrm{SL}_2(\mathbb{C})$ up to conjugacy. This recurring phenomenon is often called the *ADE correspondence*.

3 Representation Spaces and Base Change

Fix a dimension vector $\mathbf{d} = (d_i)_{i \in Q_0}$ and choose standard vector spaces

$$V_i = k^{d_i}.$$

Definition 3.1. The representation space of Q with dimension vector $\mathbf{d}$ is the affine space

$$\mathrm{Rep}(Q, \mathbf{d}) = \prod_{a:i \rightarrow j} \mathrm{Hom}_k(V_i, V_j). \quad (3.1)$$

Consequently,

$$\dim \mathrm{Rep}(Q, \mathbf{d}) = \sum_{a:i \rightarrow j} d_i d_j. \quad (3.2)$$

Define the base-change group

$$G_{\mathbf{d}} = \prod_{i \in Q_0} \mathrm{GL}(V_i). \quad (3.3)$$

For $g = (g_i)_{i \in Q_0} \in G_{\mathbf{d}}$ and $V = (V_a)_{a \in Q_1} \in \mathrm{Rep}(Q, \mathbf{d})$, set

$$g\rho(a) = g_j\rho(a)g_i^{-1} \quad \text{for every arrow } a : i \rightarrow j. \quad (3.4)$$

This is the usual change-of-basis action.

Proposition 3.2. *Two points of $\mathrm{Rep}(Q, \mathbf{d})$ lie in the same $G_{\mathbf{d}}$ -orbit if and only if the corresponding representations are isomorphic. Hence the isomorphism classes of representations with dimension vector $\mathbf{d}$ are exactly the $G_{\mathbf{d}}$ -orbits in $\mathrm{Rep}(Q, \mathbf{d})$.*

Proof. An element $g = (g_i)$ identifies two representations precisely when each g_i is an isomorphism and

$$g_j\rho(a) = \rho'(a)g_i$$

for every arrow $a : i \rightarrow j$. This is equivalent to $\rho'(a) = g_j\rho(a)g_i^{-1}$, which is exactly the orbit relation in (3.4). $\square$

4 A One-Parameter Family for $\tilde{D}_4$

The following example explains geometrically why non-Dynkin quivers may have infinitely many isomorphism classes in a fixed dimension vector.

Let Q be the quiver

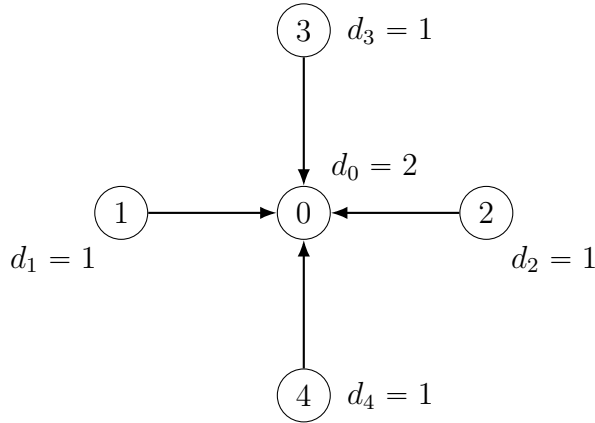


Figure 2: An orientation of the affine Dynkin graph $\tilde{D}_4$.

Thus a representation of dimension vector $(2, 1, 1, 1, 1)$ is given by four maps

$$A_i : k \longrightarrow k^2, \quad i = 1, 2, 3, 4.$$

Assume that every A_i is injective. Then

$$\ell_i = \mathrm{Im}(A_i)$$

is a line in k^2 , so the representation determines an ordered quadruple

$$(\ell_1, \ell_2, \ell_3, \ell_4) \in (\mathbb{P}^1(k))^4.$$

The factors $\mathrm{GL}_1(k)$ at the four outer vertices merely rescale the maps A_i , while the factor $\mathrm{GL}_2(k)$ at the central vertex acts simultaneously on the four lines. Therefore, isomorphism classes are ordered quadruples of points of $\mathbb{P}^1(k)$ modulo the diagonal action of $\mathrm{PGL}_2(k)$.

For $\lambda \in k \setminus \{0, 1\}$, define a representation $V(\lambda)$ by

$$\begin{aligned} A_1(1) &= (1, 0), & A_2(1) &= (0, 1), \\ A_3(1) &= (1, 1), & A_4(1) &= (\lambda, 1). \end{aligned} \tag{4.1}$$

The four image lines correspond to the ordered points

$$(\infty, 0, 1, \lambda) \in \mathbb{P}^1(k).$$

Their cross-ratio is

$$[\infty, 0; 1, \lambda] = \lambda. \tag{4.2}$$

Since the cross-ratio is invariant under $\mathrm{PGL}_2(k)$, an isomorphism

$$V(\lambda) \cong V(\mu)$$

forces $\lambda = \mu$. Because an algebraically closed field is infinite, this gives infinitely many pairwise nonisomorphic representations of the same dimension vector.

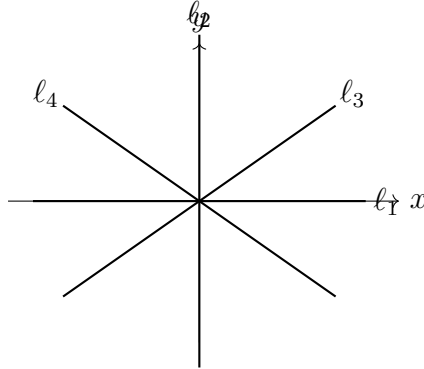


Figure 3: Four ordered lines through the origin in k^2 ; their projective configuration is measured by the cross-ratio.

5 The Tits Form and the Easy Direction of Gabriel's Theorem

We now prove the implication

$$\text{finite representation type} \implies \text{Dynkin underlying graph}.$$

The converse is the deeper direction of Gabriel's theorem.

Assume that Q has no loops. For distinct vertices $i, j \in I$, let $h_{ij} = h_{ji}$ be the number of arrows whose endpoints are i and j , with orientation forgotten. Set $h_{ii} = 0$.

Definition 5.1. Let $\{e_i\}_{i \in I}$ be the standard basis of $\mathbb{R}^I$. Define a symmetric bilinear form by

$$(e_i, e_j)_Q = \begin{cases} 2, & i = j, \\ -h_{ij}, & i \neq j. \end{cases} \quad (5.1)$$

The associated Tits quadratic form is

$$q_Q(\mathbf{x}) = \frac{1}{2}(\mathbf{x}, \mathbf{x})_Q = \sum_{i \in Q_0} x_i^2 - \sum_{\{i,j\} \subseteq Q_0} h_{ij} x_i x_j. \quad (5.2)$$

Equivalently,

$$q_Q(\mathbf{d}) = \sum_i d_i^2 - \sum_{a:i \rightarrow j} d_i d_j.$$

Proposition 5.2. Suppose that $\text{Rep}(Q, \mathbf{d})$ contains only finitely many $G_{\mathbf{d}}$ -orbits for a nonzero dimension vector $\mathbf{d}$. Then

$$q_Q(\mathbf{d}) \geq 1. \quad (5.3)$$

Proof. The affine space $\text{Rep}(Q, \mathbf{d})$ is irreducible. If it is a finite union of $G_{\mathbf{d}}$ -orbits, one of those orbits is dense. Since algebraic-group orbits are locally closed, a dense orbit is open. Let $\mathcal{O}$ be such an open orbit and choose $V \in \mathcal{O}$. Then

$$\dim \mathcal{O} = \dim G_{\mathbf{d}} - \dim (G_{\mathbf{d}})_V, \quad (5.4)$$

where $(G_{\mathbf{d}})_V$ is the stabilizer of V .

Because $\mathcal{O}$ is open in $\text{Rep}(Q, \mathbf{d})$,

$$\dim \mathcal{O} = \dim \text{Rep}(Q, \mathbf{d}) = \sum_{a:i \rightarrow j} d_i d_j.$$

Moreover,

$$\dim G_{\mathbf{d}} = \sum_i d_i^2.$$

The one-dimensional subgroup of simultaneous scalar transformations

$$\{(\lambda I_{V_i})_{i \in Q_0} : \lambda \in k^\times\}$$

fixes every representation, so

$$\dim (G_{\mathbf{d}})_V \geq 1.$$

Combining these equalities and inequalities gives

$$\sum_{a:i \rightarrow j} d_i d_j \leq \sum_i d_i^2 - 1,$$

which is equivalent to $q_Q(\mathbf{d}) \geq 1$. □

Lemma 5.3. If $q_Q(\mathbf{d}) > 0$ for every nonzero vector $\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}$, then q_Q is positive definite on $\mathbb{R}^{Q_0}$.

Proof. Suppose that q_Q is not positive definite. If there exists $\mathbf{x} \neq 0$ with $q_Q(\mathbf{x}) < 0$, then, because all off-diagonal coefficients of q_Q are nonpositive,

$$q_Q(|\mathbf{x}|) \leq q_Q(\mathbf{x}) < 0,$$

where $|\mathbf{x}| = (|x_i|)_i$. By continuity, a sufficiently close nonzero rational vector $\mathbf{y} \in \mathbb{Q}_{\geq 0}^{Q_0}$ also satisfies $q_Q(\mathbf{y}) < 0$. Clearing denominators yields a nonzero $\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}$ with $q_Q(\mathbf{d}) < 0$, a contradiction.

It remains to consider the case in which q_Q is positive semidefinite but singular. The matrix of $(\cdot, \cdot)_Q$ has integer entries and nontrivial kernel, so its kernel contains a nonzero rational vector $\mathbf{z}$. Then $q_Q(\mathbf{z}) = 0$. Again,

$$q_Q(|\mathbf{z}|) \leq q_Q(\mathbf{z}) = 0.$$

Positive semidefiniteness forces $q_Q(|\mathbf{z}|) = 0$. After clearing denominators, we obtain a nonzero vector $\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}$ satisfying $q_Q(\mathbf{d}) = 0$, again a contradiction. Therefore q_Q is positive definite. $\square$

Proposition 5.4. *Let Γ be a finite connected graph, allowing multiple edges but no loops. Its symmetric Tits form is positive definite if and only if Γ is one of the simply laced Dynkin diagrams*

$$A_n, \quad D_n, \quad E_6, \quad E_7, \quad E_8.$$

In particular, positive definiteness excludes multiple edges.

Remark 5.5. Indeed, if two vertices are joined by $m \geq 2$ edges, then the corresponding 2×2 principal submatrix is

$$\begin{pmatrix} 2 & -m \\ -m & 2 \end{pmatrix},$$

whose determinant is $4 - m^2 \leq 0$. Hence a positive-definite symmetric Tits form can only arise from a simple graph.

Proof of Theorem 2.1, (2) $\Rightarrow$ (1). Assume that, for every dimension vector $\mathbf{d}$, the variety $\text{Rep}(Q, \mathbf{d})$ has only finitely many isomorphism classes. By Proposition 3.2, it has only finitely many $G_{\mathbf{d}}$ -orbits. Proposition 5.2 therefore gives

$$q_Q(\mathbf{d}) \geq 1 \quad \text{for every nonzero } \mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}.$$

By Lemma 5.3, the Tits form q_Q is positive definite. Proposition 5.4 then implies that the underlying unoriented graph of Q is a simply laced Dynkin diagram. $\square$